Consumption of blueberries with a high-carbohydrate, low-fat breakfast decreases postprandial serum markers of oxidation.

We sought to determine whether consumption of blueberries could reduce postprandial oxidation when consumed with a typical high-carbohydrate, low-fat breakfast. Participants (n 14) received each of the three treatments over 3 weeks in a cross-over design. Treatments consisted of a high blueberry dose (75 g), a low blueberry dose (35 g) and a control (ascorbic acid and sugar content matching that of the high blueberry dose). Serum oxygen radical absorbance capacity (ORAC), serum lipoprotein oxidation (LO) and serum ascorbate, urate and glucose were measured at fasting, and at 1, 2 and 3 h after sample consumption. The mean serum ORAC was significantly higher in the 75 g group than in the control group during the first 2 h postprandially, while serum LO lag time showed a significant trend over the 3 h for both blueberry doses. Changes in serum ascorbate, urate and glucose were not significantly different among the groups. To our knowledge, this is the first report that has demonstrated that increased serum antioxidant capacity is not attributable to the fructose or ascorbate content of blueberries. In summary, a practically consumable quantity of blueberries (75 g) can provide statistically significant oxidative protection in vivo after a high-carbohydrate, low-fat breakfast. Though not tested directly, it is likely that the effects are due to phenolic compounds, either directly or indirectly, as they are a major family of compounds in blueberries with potential bioactive activity.